- Tutorial 10 -

Numerical Methods and Optimization

May 6, 2025 — ENSIA

Cutting Planes / Branch and Bound

Exercise 1: Cutting planes (Branch and Cut)

MILP (P_1):

$$\begin{cases} \max Z_1 = x_1 + 5x_2 \\ x_1 + 10x_2 \leq 20 & x_1 \in \mathbb{N} \\ x_1 \leq 2 & x_2 \in \mathbb{N} \end{cases}$$

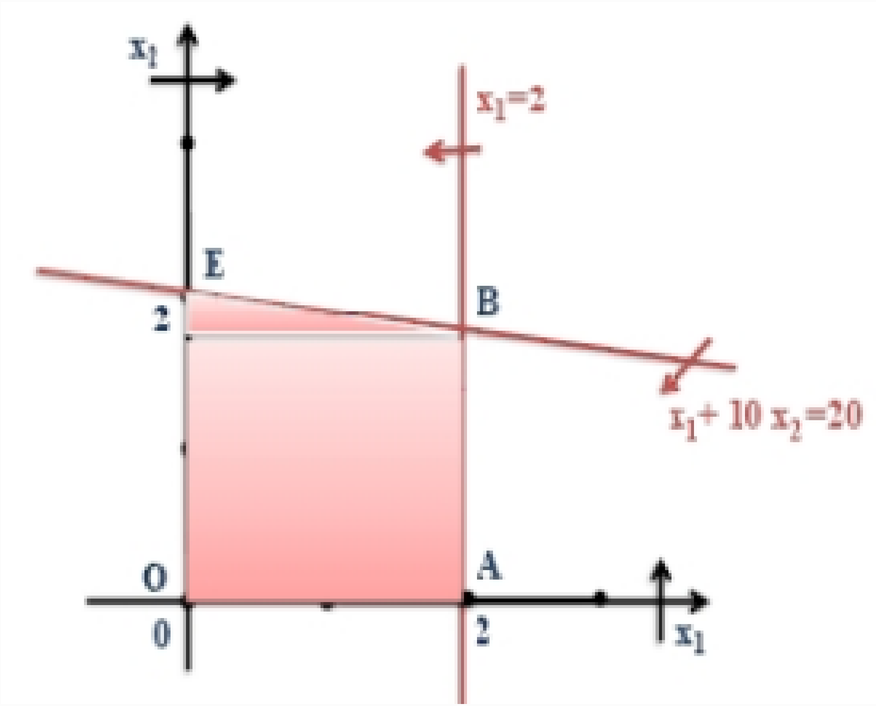
its relaxed LP (P_1') :

$$\begin{cases} \max Z_1 = x_1 + 5x_2 \\ x_1 + 10x_2 \leq 20 & x_1 \geq 0 \\ x_1 \leq 2 & x_2 \geq 0 \end{cases}$$

Step 1: Relaxe the program in R

Let S (resp. $\mathcal{D}(P_1^r)$) be the set of feasible solutions of (P_1) (resp. (P_1^r)). Clearly, $S \subseteq \mathcal{D}(P_1^r)$.
If $x^*(P_1^r)$ is the optimal solution of (P_1^r), and $\forall x \in S : Z_1(x) \leq Z_1(x^*(P_1^r))$, then $Z_1(x^*(P_1^r))$ is called an **evaluation** of S.
If, moreover, $x^*(P_1^r) \in \mathbb{N}^2$, then $Z_1(x^*(P_1^r))$ is an **exact evaluation** of S.
Otherwise, $Z_1(x^*(P_1^r))$ is a **non-exact evaluation** of S.
Evaluation of S. We evaluate S by solving the relaxed LP (P_1^r) of (P_1). We have:

$$\mathcal{D}(P_1^r) = OABE \text{ bounded convex polyhedron,}$$


x_1
$x_1=2$
E
B
2
$x_1+10\,x_2=20$
O
A
0
2
x_1

Thus, the optimum of (P_1^r) exists and is attained at least at one of the vertices of $\mathcal{D}(P_1^r)$.

Point	Z_1 value
$O(0,0)$	$Z_1(O) = 0$
$A(2,0)$	$Z_1(A) = 2$
$B(2,\frac{18}{10})$	$Z_1(B) = 11$
$E(0,2)$	$Z_1(E) = 10$

Therefore, $x^*(P_1^r) = (2, 18/10)^t$, and $Z_1^*(P_1^r) = 11$.
Since $x^*(P_1^r) = (2, 18/10)^t \notin \mathbb{N}^2$, then $Z_1^*(P_1^r) = 11$ is a **non-exact evaluation** of S $(\exists x \in S : Z_1(x) < 11)$.

Separation of S. Since $x_2^*(P_1^r) \notin \mathbb{N}$, we divide S into two subsets as follows:

$$S_1 = \left\{x \in S \mid x_2 \leq \left\lfloor \frac{18}{10} \right\rfloor\right\} \qquad S_2 = \left\{x \in S \mid x_2 \geq \left\lceil \frac{18}{10} \right\rceil\right\}$$

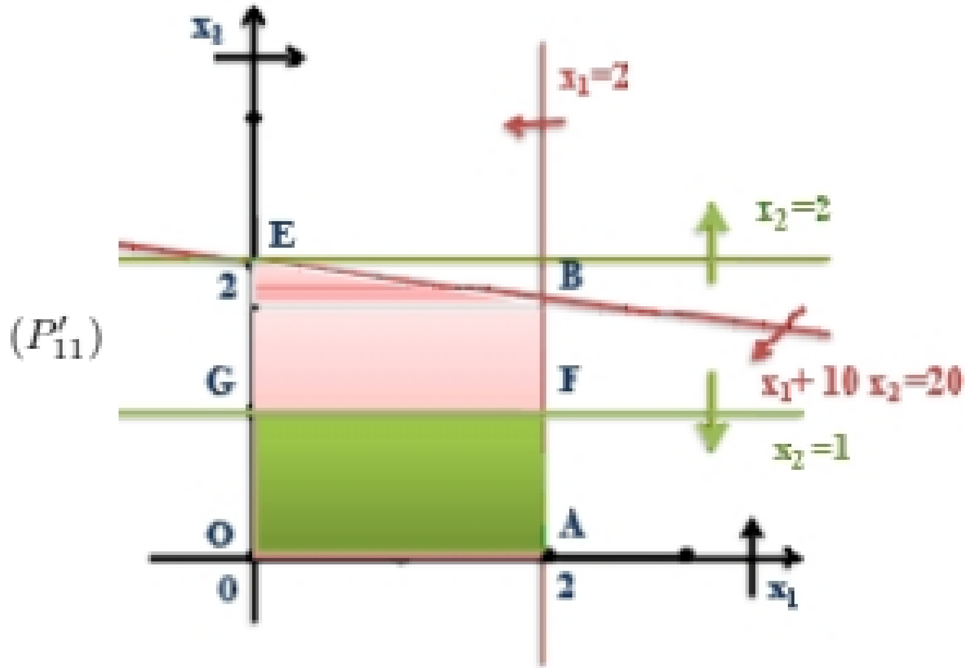
We associate with them the following two MILP problems:

$$(P_{11}) \begin{cases} \max Z_1 = x_1 + 5x_2 & \\ x_1 + 10x_2 \leq 20 & x_1 \in \mathbb{N} \\ x_1 \leq 2 & x_2 \in \mathbb{N} \\ x_2 \leq 1 & \end{cases} \qquad (P_{12}) \begin{cases} \max Z_1 = x_1 + 5x_2 & \\ x_1 + 10x_2 \leq 20 & x_1 \in \mathbb{N} \\ x_1 \leq 2 & x_2 \in \mathbb{N} \\ x_2 \geq 2 & \end{cases}$$

Evaluation of S_1 and S_2.

Graphical resolution of (P'_1) and (P'_2), the relaxed LPs of (P_1) and (P_2) respectively:

$$
(P'_1)\begin{cases}
\max Z_1 = x_1 + 5x_2 \\
x_1 + 10x_2 \leq 20 \\
x_1 \leq 2 \\
x_2 \leq 1 \\
x_1, x_2 \geq 0
\end{cases}
\qquad
(P'_2)\begin{cases}
\max Z_1 = x_1 + 5x_2 \\
x_1 + 10x_2 \leq 20 \\
x_1 \leq 2 \\
x_2 \geq 2 \\
x_1, x_2 \geq 0
\end{cases}
$$



x1
x1=2
x2=2
E
B
2
(P'11)
G
F
x1+ 10 x2=20
x2 =1
O
A
0
2
x1

$D(P_1') = OAFG$ is a bounded convex polyhedron.
The optimum of (P_1') exists and is one of its vertices.

$O(0,0)$	$A(2,0)$	$F(2,1)$	$G(0,1)$
$Z_1(O) = 0$	$Z_1(A) = 2$	$Z_1(F) = 7$	$Z_1(G) = 5$

$x^*(P_1') = (2,1)^t \in \mathbb{N}^2$, $Z_1(x^*(P_1')) = 7$: exact evaluation of S_1.

We sterilize S_1.

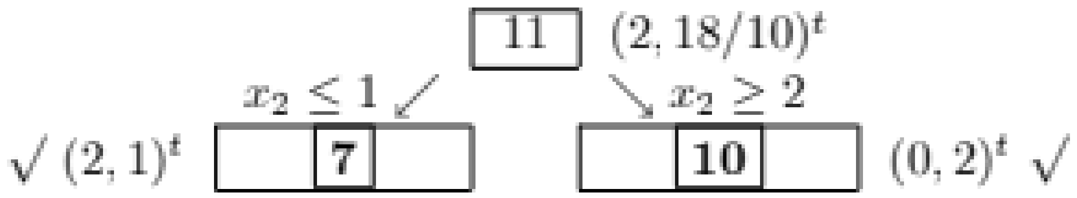
$D(P_2') = \{E\} \neq \emptyset$

The optimum of (P_2') exists and is E

$x^*_{(P_2')} = (0, 2)^t \in \mathbb{N}^2$, $Z_1(x^*_{(P_2')}) = 10$: exact evaluation of S_1.

We sterilize S_2.

According to our study, summarized in the following tree:



The optimal value of (P_1) is $\max\{7, 10\} = 10$.
So, $x^*(P_1) = (0, 2)^t$ and $Z_1^* = 10$.
Remark.

$$S = \{(0,0)_0, (1,0)_1, (2,0)_2, (0,1)_5, (1,1)_6, (2,1)_7, (0,2)_{10}\}$$

It is easy to verify that $Z_1(0, 2) = 10$ is the greatest value of Z_1 in S.

Exercise 2: Branch and Bound for the Assignment problem

1. Hungarian Method

Step 1: Row Reduction

Subtract the minimum element of each row from every element in the row.

$$\begin{bmatrix} 9 & 2 & 7 & 8 \\ 6 & 4 & 3 & 7 \\ 5 & 8 & 1 & 8 \\ 7 & 6 & 9 & 4 \end{bmatrix} \rightarrow \begin{bmatrix} 7 & 0 & 5 & 6 \\ 3 & 1 & 0 & 4 \\ 4 & 7 & 0 & 7 \\ 3 & 2 & 5 & 0 \end{bmatrix}$$

Step 2: Column Reduction

Subtract the minimum element of each column from every element in the column.

$$\begin{bmatrix} 7 & 0 & 5 & 6 \\ 3 & 1 & 0 & 4 \\ 4 & 7 & 0 & 7 \\ 3 & 2 & 5 & 0 \end{bmatrix} \rightarrow \begin{bmatrix} 4 & 0 & 5 & 6 \\ 0 & 1 & 0 & 4 \\ 1 & 7 & 0 & 7 \\ 0 & 2 & 5 & 0 \end{bmatrix}$$

W1 do J2,

Step 3: Assignment

We try to assign 0s in such a way that no row or column is used more than once:

- Row 1: assign to column 2 (0) - Row 2: assign to column 1 (0) - Row 3: assign to column 3 (0) - Row 4: assign to column 4 (0)

Assignment:

Worker 1 → Job 2

Worker 2 → Job 1

Worker 3 → Job 3

Worker 4 → Job 4

Total Cost:

$$c_{12} + c_{21} + c_{33} + c_{44} = 2 + 6 + 1 + 4 = \boxed{13}$$

2. Greedy Method

Choose the smallest available cost in the matrix step by step, ensuring no repeated row or column:

1. Smallest value: 1 at $(3, 3)$. Assign Worker 3 to Job 3.
2. Remove row 3 and column 3.
3. Next smallest: 2 at $(1, 2)$. Assign Worker 1 to Job 2.
4. Remove row 1 and column 2.
5. Remaining matrix:

$$\begin{bmatrix} - & - & - & - \\ 6 & - & - & 7 \\ - & - & - & 4 \end{bmatrix}$$

6. Smallest: 4 at $(4, 4)$. Assign Worker 4 to Job 4.
7. Remaining: Worker 2 and Job 1. Assign directly.

Greedy Assignment:

Worker 3 $\rightarrow$ Job 3

Worker 1 $\rightarrow$ Job 2

Worker 4 $\rightarrow$ Job 4

Worker 2 $\rightarrow$ Job 1

Total Cost:

$$c_{33} + c_{12} + c_{44} + c_{21} = 1 + 2 + 4 + 6 = \boxed{13}$$

3. Branch and Bound

In both Breadth-First Search (BFS) and Depth-First Search (DFS), the selection of the next node to explore is blind—that is, it does not prioritize nodes that are more likely to lead to an optimal solution. These uninformed strategies treat all nodes equally, without considering their potential to reach the goal efficiently.

To overcome this limitation, we can use an informed search strategy that leverages an intelligent ranking function, often referred to as an approximate cost function. This function helps guide the search process by assigning a cost to each live node, enabling the algorithm to avoid exploring subtrees unlikely to contain optimal solutions. This approach is similar to BFS, but instead of following the strict FIFO order, it selects the live node with the lowest estimated cost.

Evaluation:

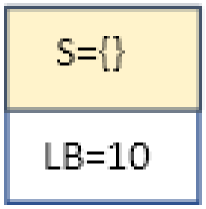
There are two common approaches to estimate the cost function:

Row-based Minimization: For each worker (row), select the minimum cost job from the list of unassigned jobs (i.e., take the minimum entry from each row).

Column-based Minimization: For each job (column), choose the worker with the lowest cost from the list of unassigned workers (i.e., take the minimum entry from each column).

In this exercise, the first approach is followed.
Let's calculate the **total lower bound** if the problem (it is a minimization problem).
We take the minimum entry from each row: 2 + 3 + 1 + 4 = 10

	Job 1	Job 2	Job 3	Job 4
A	9	$\boxed{2}$	7	8
B	6	4	$\boxed{3}$	7
C	5	8	$\boxed{1}$	8
D	7	6	9	$\boxed{4}$



Level 1 : Worker A

A → 1: $LB = 9 + 3 + 1 + 4 = 17$

	Job 1	Job 2	Job 3	Job 4
A	9			
B		4	3	7
C		8	1	8
D		6	9	4

A → 2: $LB = 2 + 3 + 1 + 4 = 10$

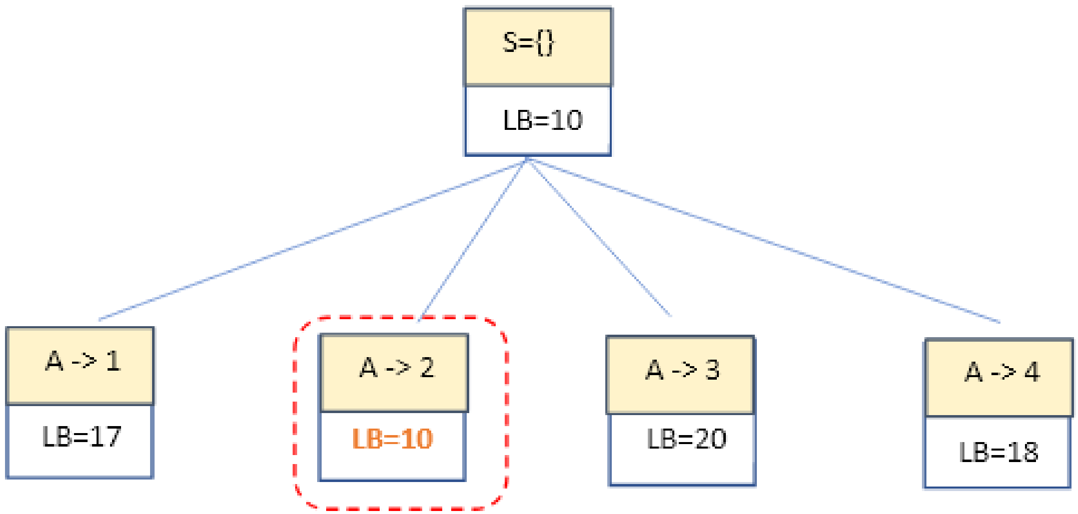
	Job 1	Job 2	Job 3	Job 4
A		$\boxed{2}$		
B	6		$\boxed{3}$	7
C	5		$\boxed{1}$	8
D	7		9	$\boxed{4}$

A → 3: $LB = 7 + 4 + 5 + 4 = 20$

	Job 1	Job 2	Job 3	Job 4
A			7	
B	6	4		7
C	5	8		8
D	7	6		4

A → 4: $LB = 8 + 3 + 1 + 6 = 18$

	Job 1	Job 2	Job 3	Job 4
A				$\boxed{8}$
B	6	4	$\boxed{3}$	
C	5	8	$\boxed{1}$	
D	7	$\boxed{6}$	9	


S={}
LB=10
A -> 1
LB=17
A -> 2
LB=10
A -> 3
LB=20
A -> 4
LB=18

Level 2 : Worker B

B → 1: $LB = 2 + 6 + 1 + 4 = 13$

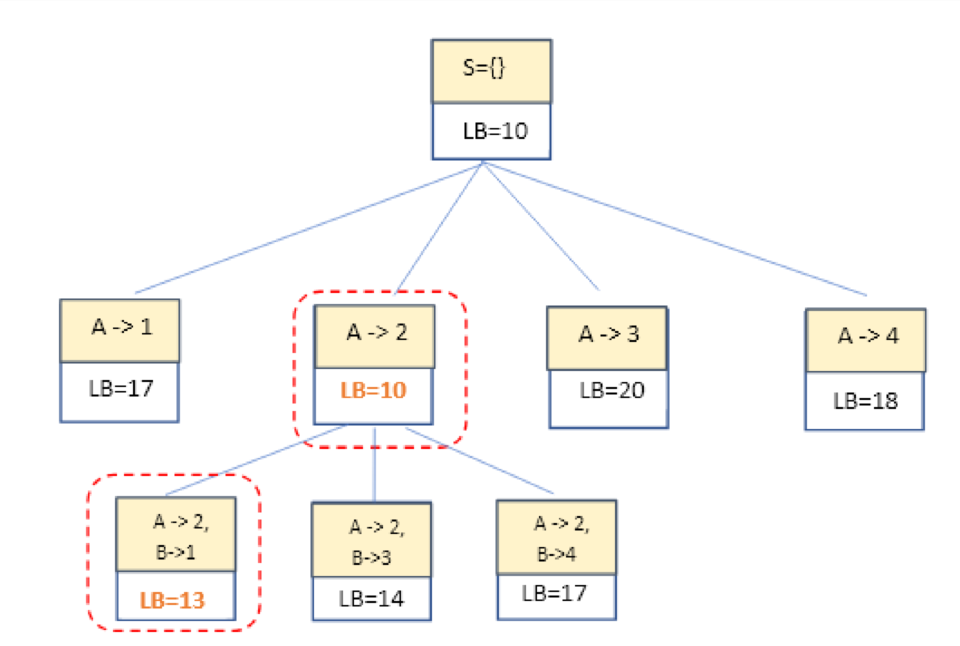
	Job 1	Job 2	Job 3	Job 4
A		2		
B	6			
C			1	8
D			9	4

B → 3: $LB = 2 + 3 + 5 + 4 = 14$

	Job 1	Job 2	Job 3	Job 4
A		2		
B			3	
C	5			8
D	7			4

B → 4: $LB = 2 + 7 + 1 + 7 = 17$

	Job 1	Job 2	Job 3	Job 4
A		2		
B				7
C	5		1	
D	7		9	


S={}
LB=10
A -> 1
LB=17
A -> 2
LB=10
A -> 3
LB=20
A -> 4
LB=18
A -> 2,
B->1
LB=13
A -> 2,
B->3
LB=14
A -> 2,
B->4
LB=17

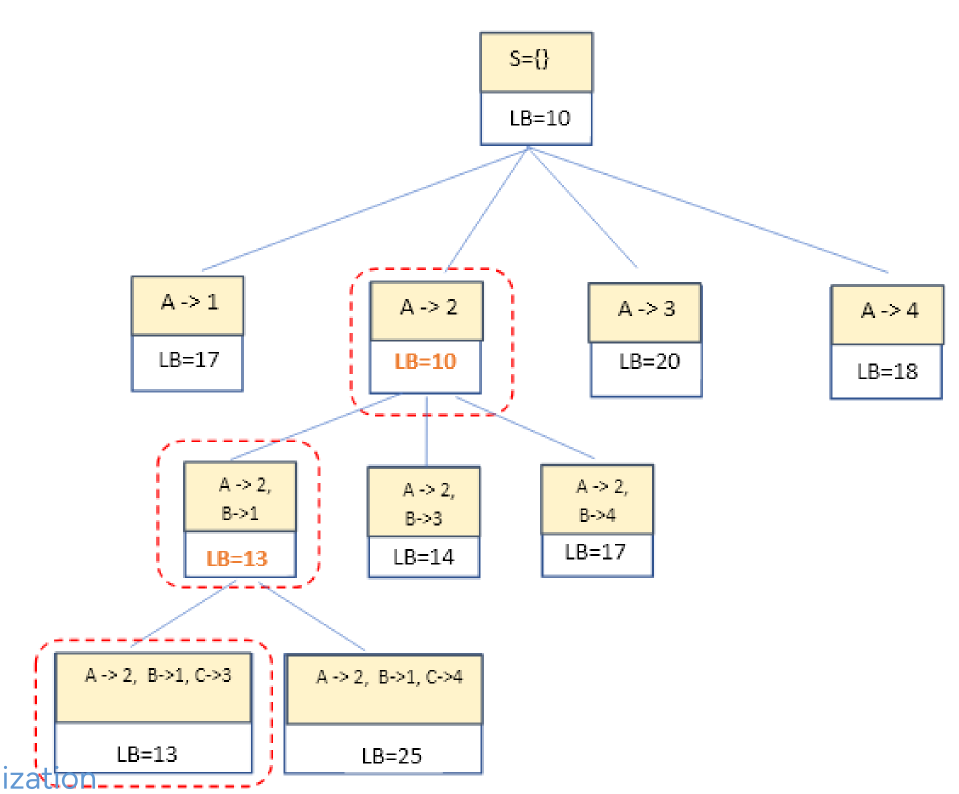
Level 3 : Worker C

C → 3: $LB = 2 + 6 + 1 + 4 = 13$

	Job 1	Job 2	Job 3	Job 4
A		2		
B	6			
C			1	
D				4

C → 4: $LB = 2 + 6 + 8 + 9 = 25$

	Job 1	Job 2	Job 3	Job 4
A		2		
B	6			
C				8
D			9	


S={}
LB=10
A -> 1
LB=17
A -> 2
LB=10
A -> 3
LB=20
A -> 4
LB=18
A -> 2,
B->1
LB=13
A -> 2,
B->3
LB=14
A -> 2,
B->4
LB=17
A -> 2, B->1, C->3
LB=13
A -> 2, B->1, C->4
LB=25

Level 4 : Worker D

D → 4: $LB = 2 + 6 + 1 + 4 = 13$

At this stage of the Branch and Bound process:

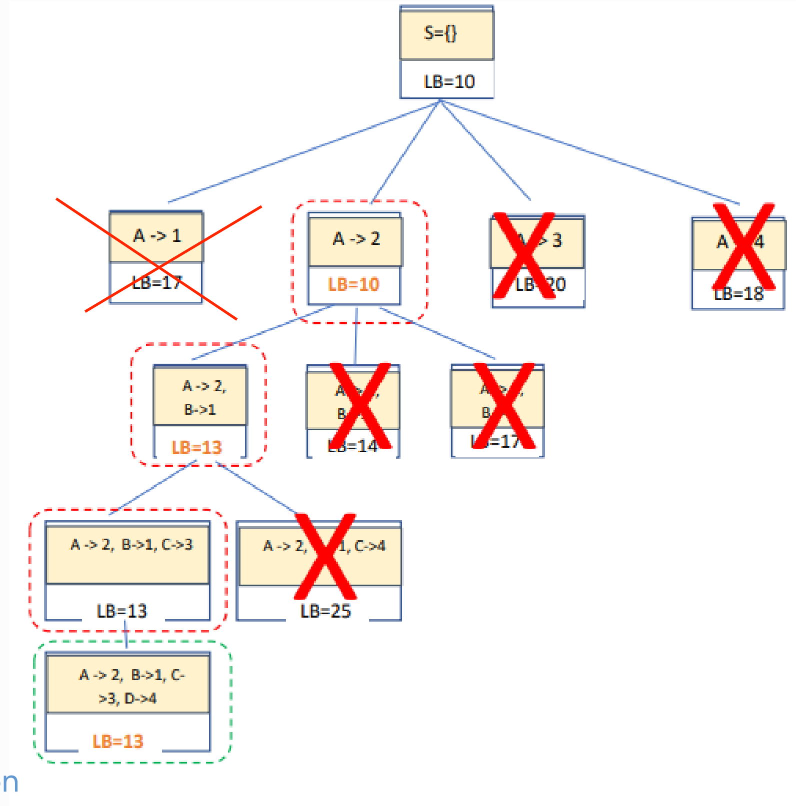
- **This node is a leaf** because we have obtained a **complete solution** (i.e., all assignments have been made).
- The **lower bound (LB)** at this node corresponds to the **exact cost** of the complete assignment.
- We then **sterilize all nodes** in the search tree whose evaluations are greater than or equal to 13, as they are considered **non-promising nodes**.
- After sterilization, we observe that **no more nodes remain** to branch and explore.
- Therefore, we **terminate the Branch and Bound algorithm**.

The optimal assignment is:

$$\begin{aligned}\text{Worker A} &\rightarrow \text{Job 2},\\ \text{Worker B} &\rightarrow \text{Job 1},\\ \text{Worker C} &\rightarrow \text{Job 3},\\ \text{Worker D} &\rightarrow \text{Job 4}.\end{aligned}$$

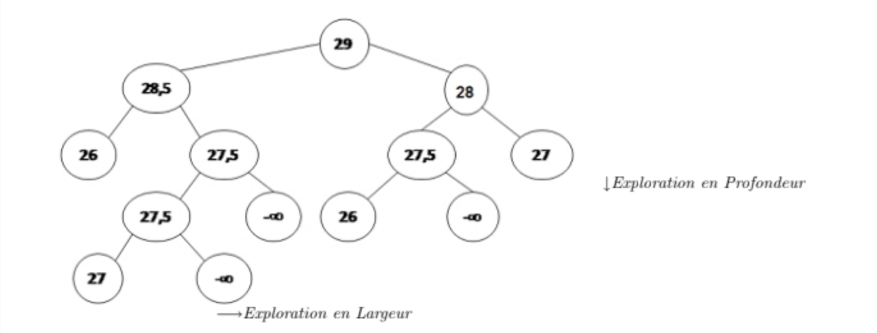
Total Cost: $\boxed{13}$

Cutting Planes / Branch and Bound

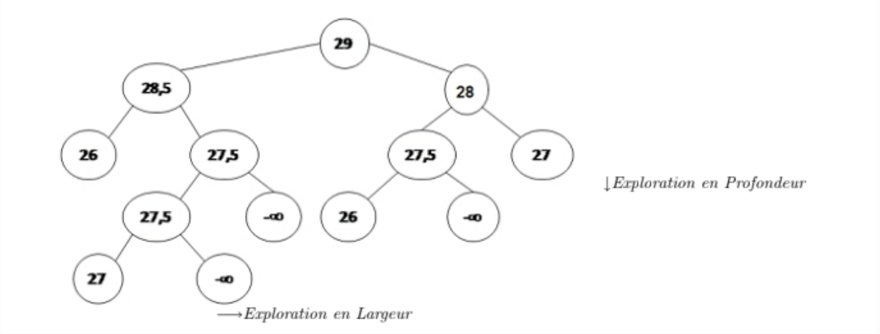


Exercise 3: Analysis of a Branch-and-Bound Search Tree

a) Since the evaluations decrease when traversing the tree in depth, this is a maximization problem. We also note that the optimal solutions are given by the nodes with exact evaluation 27.



b) A depth-first search leads us to the value 27 from the evaluation set 27, 5 first. We do not move to another level until all nodes at the current level have been completely fathomed.



c) A breadth-first search yields the exact evaluation 27 from the node with evaluation 28 first. We move from left to right and do not proceed to another level until the current level has been fully explored.